Signals and systems IIA

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1. Course Syllabus

1. Introduction to Signals and systems
   - Basic properties of signals and systems.
2. Continuous-Time Fourier Series
   - Definition of exponential Fourier Series.
   - Fourier Series analysis of Linear Time Invariant Continuous-Time Systems
3. Continuous-Time Fourier Transform
   - Definition of the Fourier Integral Transform
   - Fourier Transform analysis of Linear Time Invariant Continuous-Time Systems
4. Butterworth Filter Design
   - Low-Pass Filter design
   - High-Pass Filter design
   - Band-Pass Filter design
   - Band-Stop Filter design
5. Nyquist Sampling Theorem
   - Nuyquist Sampling Theorem for band-limited signals
6. State-Space Analysis
   - Deriving State-Space equations for *RLC*
   - Solution of State-Space equations
   - Transfer Function $\longleftrightarrow$ State-Space representation
   - Linear transformation of state vector
   - Controllability and Observability

2. Introduction to Signals and systems

A **signal** is a function of one or more independent variables that carries information about a physical phenomenon. Common electrical engineering examples include the voltage across a component, current through a conductor, acoustic pressure recorded by a microphone, and the ECG trace of cardiac activity. Throughout this course the independent variable is time $t \in \mathbb{R}$ and we study **continuous-time (CT)** signals — functions defined for every instant t, as opposed to **discrete-time (DT)** signals which are defined only at integer time steps.

2.1. Signal properties

2.1.1. Periodicity

A signal $x(t)$ is periodic with period T if and only if

$$x(t) = x(t + nT), \text{for } n \in \mathbb{N} \tag{1}$$

.

2.1.2. Even/Odd signals

The signal $x(t)$ is said to be even if $x(t) = x(-t)$ and odd if $-x(t) = x(-t)$. Any signal $x(t)$ can be decomposed into an even and odd part $x_e(t)$ and $x_o(t)$

$$\begin{aligned} x(t) = &\quad x_e(t) + x_o(t) \\ \text{where } x_e(t) = &\frac{1}{2}(x(t) + x(-t)) \\ x_o(t) = &\frac{1}{2}(x(t) - x(-t)) \end{aligned} \tag{2}$$

.

It can easily be shown that odd(even) signals have the following properties.

1. The graph of an even signals is symmetric about the vertical axis.
2. The graph of an odd signal has 180^0 rotational symmetry.
3. The product of two even signals is an even signal.
4. The product of two odd signals is an even signals.
5. The product of an odd signal and even signal is an odd signal.

Decomposing an signal into it's odd and even parts is often useful in simplifying integrals because of the following properties.

$$\begin{aligned} \int_{-a}^{a} x_o(t)\,\mathrm{d}t = &\quad 0 \\ \int_{-a}^{a} x_e(t)\,\mathrm{d}t = &\quad 2\int_{0}^{a} x_{e(t)}\,\mathrm{d}t \end{aligned} \tag{3}$$

2.1.3. Signal energy

The energy of a signal $x(t)$ denoted E_x is defined as

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, \mathrm{d}t. \tag{4}$$

E_x can be understood as a measure of the size of $x(t)$. The signal $x(t)$ is an *energy* signal if it has finite energy, a necessary condition for the energy to be finitie is that the signal amplitude $\longrightarrow 0$ as $|t| \longrightarrow \infty$.

2.1.4. Signal power

The power of a signal $x(t)$ denoted P_x is defined as

$$P_x = \lim_{T \to \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 \, \mathrm{d}t. \tag{5}$$

P_x is the time average of the signal amplitude squared i.e $\sqrt{P_x}$ is the root-mean-square(**rms**) value of $x(t)$. The signal $x(t)$ is a *power* signal if it has finite power.

> **Note**
>
> Signal power involves averaging energy over an infinite duration which means a signal with finite energy has zero power, and a signal with finite power has infinite energy. Therefore a signal cannot be both and energy signal and a power signal. There exists signals that are neither energy signals nor power signals for example ramp signals.

2.1.5. Deterministic and random signals

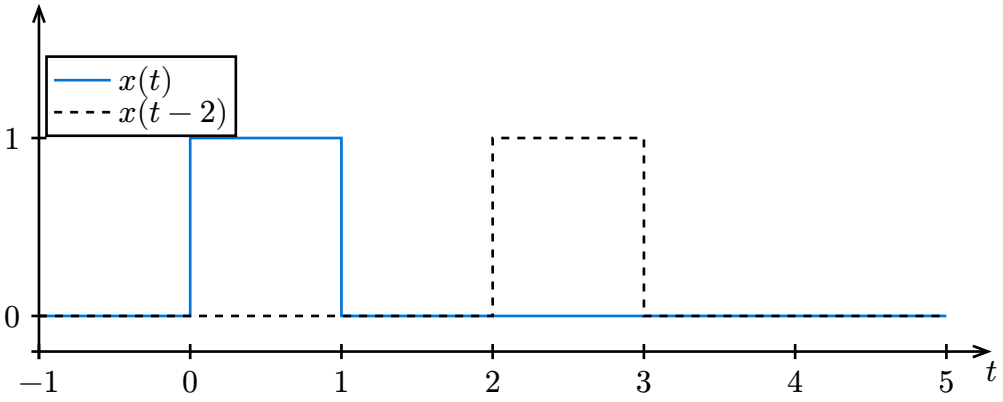
A **deterministic** signal is completely specified by a mathematical expression — its values are known precisely for all time. A **random** (or stochastic) signal takes values that cannot be predicted in advance and must be characterised statistically. This course deals exclusively with deterministic signals.

2.2. Signal operations

Three fundamental operations on the independent variable t transform a signal in time. Understanding these is essential for working with convolution and Fourier analysis.

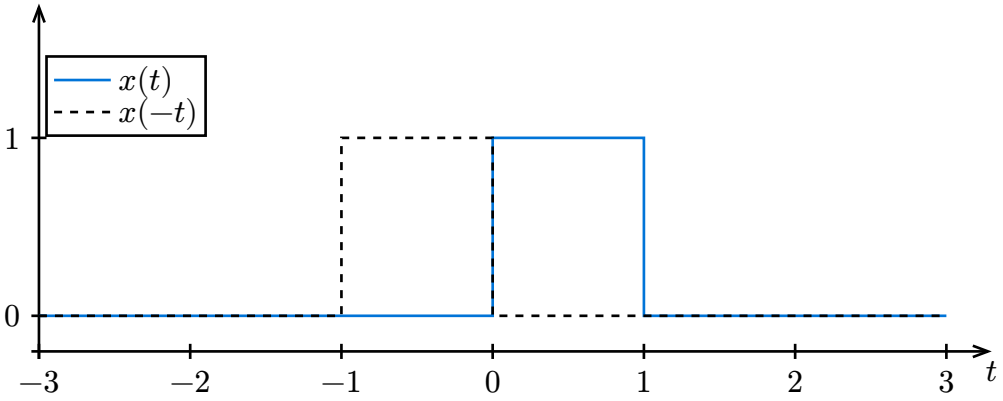
2.2.1. Time shifting

Given $x(t)$, the **time-shifted** signal is $x(t - t_0)$. It delays the signal by t_0 seconds when $t_0 > 0$ and advances it by $|t_0|$ seconds when $t_0 < 0$. Every feature of $x(t)$ moves by exactly t_0 along the time axis.



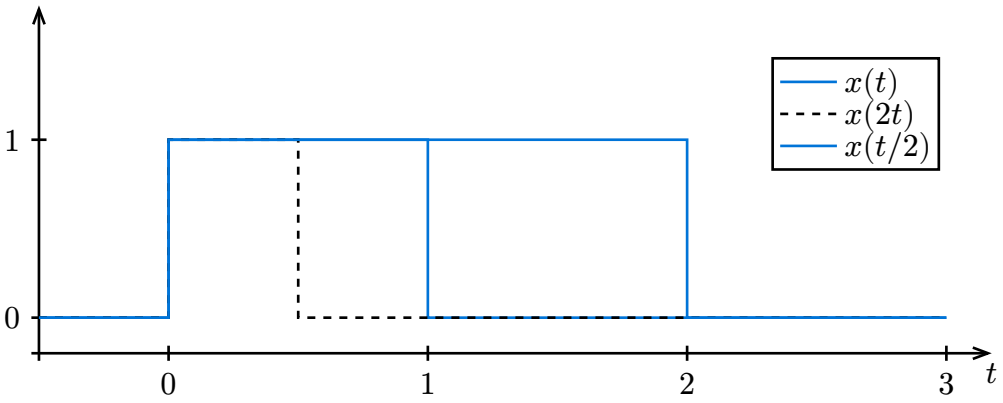
2.2.2. Time reversal

The **time-reversed** signal is $x(-t)$, the reflection of $x(t)$ about the vertical axis $t = 0$. A feature that occurs at time t_1 in $x(t)$ appears at $-t_1$ in $x(-t)$.



2.2.3. Time scaling

The **time-scaled** signal is $x(at)$ where $a > 0$. When $a > 1$ the signal is **compressed** in time; when $0 < a < 1$ it is **expanded**. A feature at time t_1 in $x(t)$ moves to $\frac{t_1}{a}$ in $x(at)$.



2.2.4. Combined transformations

When operations are combined the order matters. To evaluate $x(at - t_0)$ from $x(t)$, factor the argument:

$$x(at - t_0) = x\left(a\left(t - \frac{t_0}{a}\right)\right). \tag{6}$$

The procedure is then: (1) time-shift $x(t)$ right by $\frac{t_0}{a}$ to obtain $x\left(t-\frac{t_0}{a}\right)$, then (2) time-scale by a to obtain $x\left(a\left(t-\frac{t_0}{a}\right)\right) = x(at-t_0)$.

Example: Obtain $y(t) = x(2t+3)$ from $x(t)$. Rewrite as $x\left(2\left(t+\frac{3}{2}\right)\right)$. Shift $x(t)$ left by $\frac{3}{2}$ to get $x\left(t+\frac{3}{2}\right)$, then compress by factor 2 to get $x\left(2\left(t+\frac{3}{2}\right)\right) = x(2t+3)$.

2.3. System properties

2.3.1. Linearity

Consider a system $\boldsymbol{H}$ such that $\boldsymbol{H}(x(t)) = y(t)$ where $x(t)$ is the input and $y(t)$ is the output. The system $\boldsymbol{H}$ is said to be linear if had the following properties:

- **homogeneity/scaling property** : For $\boldsymbol{k}$ a complex number $\boldsymbol{H}(\boldsymbol{k}x(t)) = \boldsymbol{k}\boldsymbol{H}(x(t)) = \boldsymbol{k}y(t)$.
- **superposition property**: Let the response of system $\boldsymbol{H}$ to inputs $x_1(t)$ and $x_2(t)$ be $y_1(t) = \boldsymbol{H}(x_1(t))$ and $y_2(t) = \boldsymbol{H}(x_2(t))$ then $\boldsymbol{H}(x_1(t) + x_2(t)) = y_1(t) + y_2(t)$.

2.3.2. Time invariance

Consider a system $\boldsymbol{H}$ such that $\boldsymbol{H}(x(t)) = y(t)$ where $x(t)$ is the input and $y(t)$ is the output. $\boldsymbol{H}$ is said to be time invariant if $\boldsymbol{H}(x(t-t_0)) = y(t-t_0), t_0 \in \mathbb{R}$. Time shifting the input by t_0 shifts the output by t_0.

2.3.3. Memory

A system $\boldsymbol{H}$ is said to have memory if the output at time t_0 i.e. $y(t_0)$ is a function of/depends on input values other that $x(t_0)$.

2.3.4. Invertibility

A system $\boldsymbol{H}$ is invertible if distinct input results in distinct outputs. For an invertible system $\boldsymbol{H}$ the exists a system $\boldsymbol{H}^{-1}$ it's inverse such that $\boldsymbol{H}^{-1}(\boldsymbol{H}(x(t))) = x(t)$.

2.3.5. Causality

The system $\boldsymbol{H}$ is causal if the system output at anytime t_0 dependence only on input values for $t \leq t_0$

2.3.6. Bounded Input Bounded Output(BIBO) stability

A system $\boldsymbol{H}$ is said to be BIBO stable if the output $y(t)$ remains bounded for any bounded input. More precisely for $x(t)$ such that $|x(t)| \leq M$ where $M > 0$, there exists $R > 0$ such that $|\boldsymbol{H}(x(t))| \leq R$ for all t.

2.3.7. Worked example: system classification

Consider the system $\boldsymbol{H}$ defined by $y(t) = t \cdot x(t)$. Classify it against all six properties.

Linearity: For inputs $x_1(t)$ and $x_2(t)$ with responses $y_1(t) = tx_1(t)$ and $y_2(t) = tx_2(t)$,

$$\boldsymbol{H}(k_1x_1(t) + k_2x_2(t)) = t(k_1x_1(t) + k_2x_2(t)) = k_1y_1(t) + k_2y_2(t). \tag{7}$$

The system is **linear**. ✓

Time invariance: Shift the input to $x_1(t) = x(t - t_0)$, giving $y_1(t) = tx(t - t_0)$. But the delayed output is $y(t - t_0) = (t - t_0)x(t - t_0) \neq y_1(t)$. The system is **time varying**. $\times$

Memory: $y(t_0) = t_0 x(t_0)$ depends only on $x(t_0)$, so the system is **memoryless**. ✓

Causality: Since $y(t_0)$ depends only on $x(t_0)$, the system is **causal**. ✓

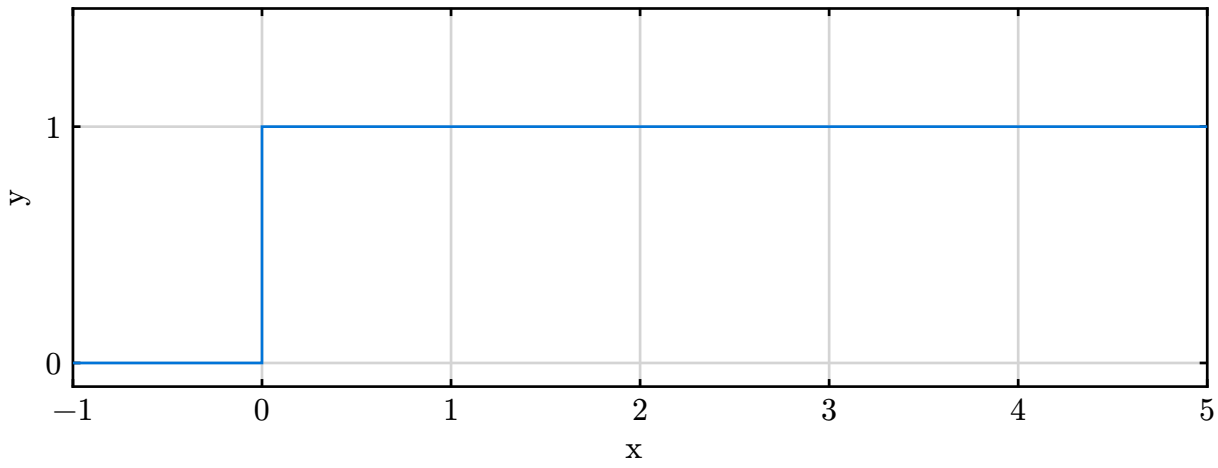
Invertibility: $x(t) = \frac{y(t)}{t}$ for $t \neq 0$, but the inverse is undefined at $t = 0$. The system is **not invertible**. $\times$

BIBO stability: Suppose $|x(t)| \leq M$ for all t. Then $|y(t)| = |t|\ |x(t)| \leq M|t|$, which is unbounded as $|t| \to \infty$. The system is **not BIBO stable**. $\times$

2.4. Signal Models

2.4.1. Unit step function

$$u(t) = \begin{cases} 1 \text{ if } t > 0 \\ 0 \text{ if } t < 0 \end{cases} \tag{8}$$

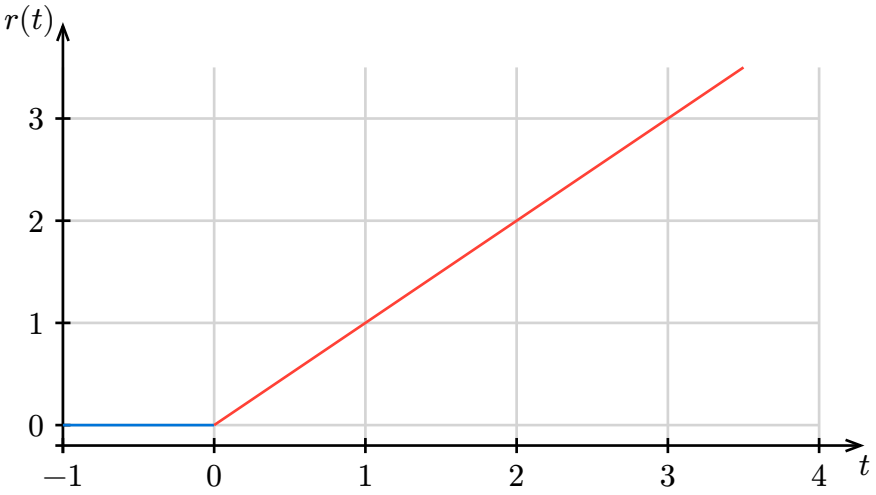


2.4.2. Unit ramp function

The unit ramp $r(t)$ is defined as

$$r(t) = \begin{cases} t \text{ if } t \geq 0 \\ 0 \text{ if } t < 0 \end{cases} \tag{9}$$

and can be written compactly as $r(t) = t \cdot u(t)$. It is neither an energy signal nor a power signal.



The unit ramp is the integral of the unit step:

$$r(t) = \int_{-\infty}^{t} u(\tau)\,\mathrm{d}\tau, \quad \frac{\mathrm{d}r(t)}{\mathrm{d}t} = u(t). \tag{10}$$

2.4.3. Unit Impulse function

The unit impulse denoted $\delta(t)$ is the function defined by the following equations

$$\begin{aligned} \delta(t - t_0) &= 0, t \neq t_0 \\ \int_{-\infty}^{\infty} \delta(t - t_0)\,\mathrm{d}t &= 1 \end{aligned} \tag{11}$$

The impulse function has a useful property, the sifting property that is

$$\int_{-\infty}^{\infty} f(t)\delta(t - t_0)\,\mathrm{d}t = f(t_0) \tag{12}$$

$$f(t)\delta(t - t_0) = f(t_0)\delta(t - t_0) \tag{13}$$

The unit impulse and unit step are related by

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\,\mathrm{d}\tau, \quad \delta(t) = \frac{\mathrm{d}u(t)}{\mathrm{d}t}. \tag{14}$$

The impulse also satisfies the **scaling property**

$$\delta(at) = \frac{1}{|a|}\delta(t), \quad a \neq 0, \tag{15}$$

which follows from requiring that $\int_{-\infty}^{\infty} \delta(at)\,\mathrm{d}t = 1$.

2.5. Complex exponential signal

The complex exponential signal $x(t)$ is defined as

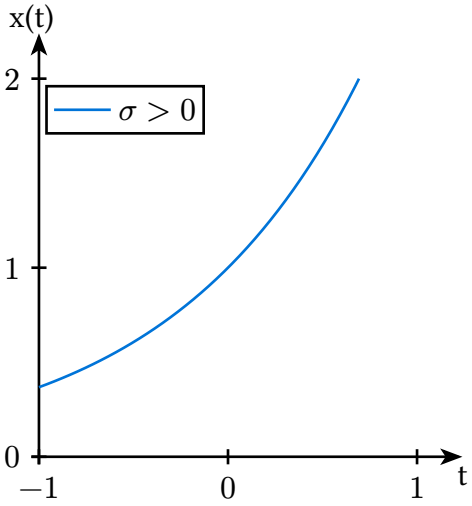
$$x(t) = e^{st}, s = \sigma + j\omega \tag{16}$$

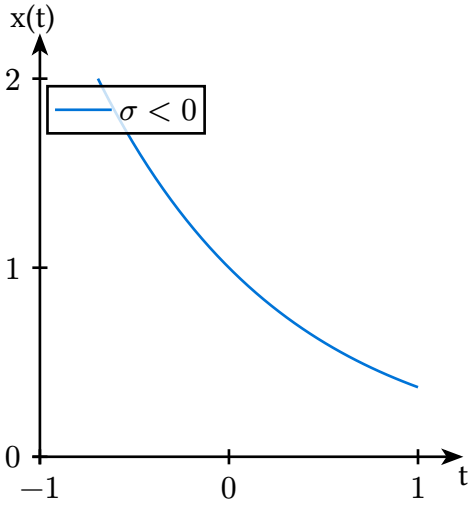
which encompases a large variety of signals.

1. Case when $s = 0$

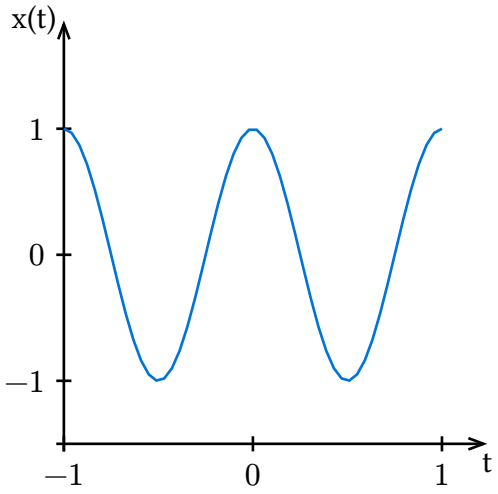
$x(t) = 1$ i.e. models constant signals.

2. Case when s is pure real i.e. $\omega = 0, x(t) = e^{\sigma t}$.

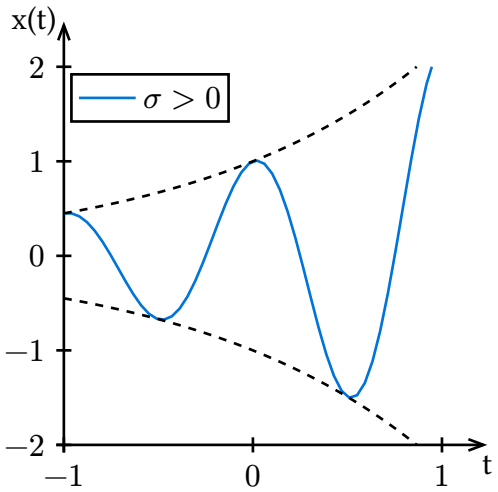


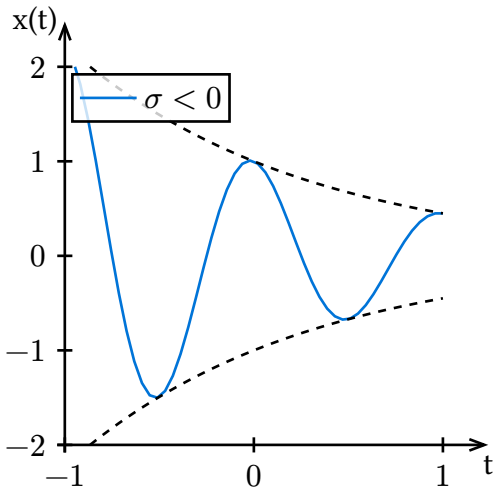


3. Case when s is pure imaginary i.e $\sigma = 0, x(t) = e^{j\omega t}$.



4. Case when s is complex i.e $x(t) = e^{(\sigma + j\omega)t} = e^{\sigma t} e^{j\omega t}$.





By Euler's formula

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t), \tag{17}$$

the complex exponential with pure imaginary exponent $s = j\omega$ decomposes into cosine (real part) and sine (imaginary part) components. This leads directly to the real sinusoidal signal.

2.5.1. Sinusoidal signal

The **sinusoidal signal** is defined as

$$x(t) = A\cos(\omega t + \varphi) \tag{18}$$

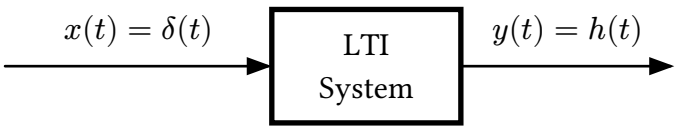
where A is the **amplitude**, ω is the **angular frequency** (rad/s), and φ is the **phase** (rad). The fundamental period is $T = 2\frac{\pi}{\omega}$ and the cyclic frequency is $f = \frac{1}{T} = \frac{\omega}{2\pi}$ Hz. Any real sinusoid can be expressed as the real part of a complex exponential:

$$A\cos(\omega t + \varphi) = A\ \mathrm{Re}\left\{e^{j(\omega t+\varphi)}\right\}. \tag{19}$$

2.6. Time domain system analysis(refresher)

2.6.1. Impulse response representation

The impulse response function of an LTI system denoted as $h(t)$ is defined as the system's response to the unit impulse function $\delta(t)$.



The system response to any input $x(t)$ can be expressed as an integral involving the input and the system impulse response function $h(t)$,

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau = x(t) * h(t). \tag{20}$$

This integral is called the convolution integral.

2.6.1.1. System properties

1. **Memoryless systems** : An LTI system is memoryless if and only if $h(t) = K\delta(t)$, that is $y(t) = Kx(t)$.
2. **Invertibility** : An LTI system with impulse response $h(t)$ is invertible only if the function $h_i(t)$ can be found such that $h(t) * h_i(t) = \delta(t)$.
3. **Causality** : An LTI system with impulse response $h(t)$ is causal if $h(t) = 0$ for all $t < 0$.
4. **Stability** : An LTI system with impulse response $h(t)$ is BIBO stable if

$$\int_{-\infty}^{\infty} |h(t)|\ dt < \infty. \tag{21}$$

If $h(t)$ satisfies this condition it is said to be absolutely integrable.

2.6.2. Worked convolution example

Compute $y(t) = x(t) * h(t)$ where

$$x(t) = u(t) - u(t-2), \quad h(t) = e^{-t}u(t). \tag{22}$$

$x(t)$ is a rectangular pulse of unit amplitude and duration 2 s; $h(t)$ is a causal exponential decay with time constant 1 s.

Using the convolution integral with $x(\tau) = 1$ only for $0 \le \tau \le 2$:

$$y(t) = \int_0^2 e^{-(t-\tau)}u(t-\tau)\,\mathrm{d}\tau. \tag{23}$$

Case 1, $t < 0$: $u(t-\tau) = 0$ for all $\tau \in [0, 2]$, so $y(t) = 0$.

Case 2, $0 \le t < 2$: The step $u(t-\tau)$ is nonzero only for $\tau \le t$, so

$$y(t) = \int_0^t e^{-(t-\tau)}\,\mathrm{d}\tau = e^{-t}[e^{\tau}]_0^t = 1 - e^{-t}. \tag{24}$$

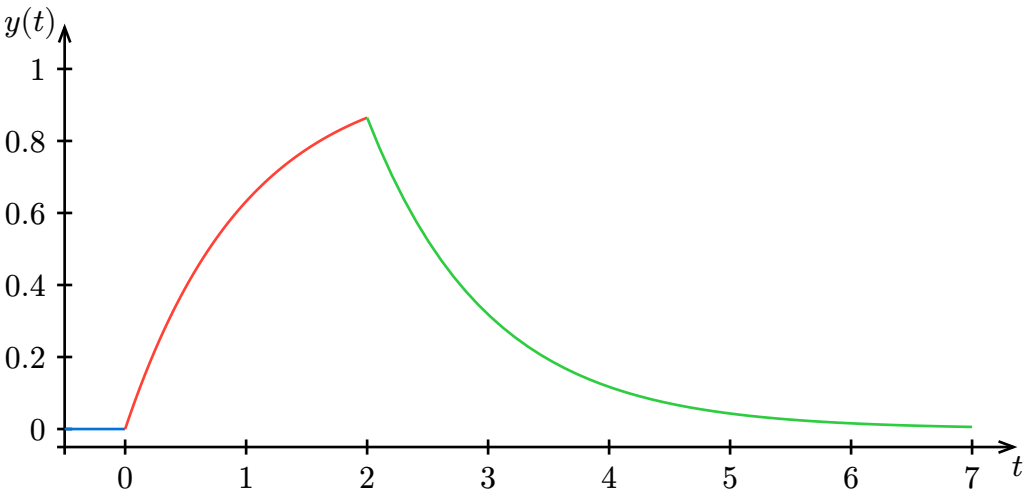
Case 3, $t \ge 2$: The step is nonzero for all $\tau \in [0, 2]$, so

$$y(t) = \int_0^2 e^{-(t-\tau)}\,\mathrm{d}\tau = e^{-t}[e^{\tau}]_0^2 = (e^2 - 1)e^{-t}. \tag{25}$$

The complete output is therefore

$$y(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 - e^{-t} & \text{if } 0 \le t < 2 \\ (e^2 - 1)e^{-t} & \text{if } t \ge 2. \end{cases} \tag{26}$$

Note that $y(t)$ is continuous at $t = 2$: both expressions give $1 - e^{-2} \approx 0.865$ there. The output rises from zero, reaches its peak at $t = 2$, and then decays exponentially — the hallmark of an RC-type response to a pulse input.



3. Continuous-Time Fourier Series

3.1. Motivation: Eigenfunctions of LTI Systems

Recall from Chapter 1 that the output of an LTI system with impulse response $h(t)$ is the convolution $y(t) = x(t) * h(t)$. Suppose the input is the complex exponential $x(t) = e^{j\omega t}$. Then

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)}\,\mathrm{d}\tau \\ &= e^{j\omega t}\int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau}\,\mathrm{d}\tau \\ &= H(j\omega)e^{j\omega t}, \end{aligned} \tag{27}$$

where

$$H(j\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau}\,\mathrm{d}\tau \tag{28}$$

is a complex number depending only on ω. The output is simply the input $e^{j\omega t}$ scaled by $H(j\omega)$: the complex exponential passes through an LTI system **unchanged in shape**, multiplied only by the complex constant $H(j\omega)$. Signals with this property are called **eigenfunctions** of the system, and $H(j\omega)$ is the corresponding **eigenvalue**.

> **Key insight**
>
> If a periodic signal $x(t)$ can be written as a sum of complex exponentials, then the LTI output $y(t)$ is the same sum with each exponential scaled by $H(j\omega)$ at that frequency. This is the central motivation for the Fourier Series.

3.2. Harmonically Related Complex Exponentials

A signal $x(t)$ is periodic with period T_0 if $x(t) = x(t + T_0)$ for all t. The **fundamental angular frequency** is $\omega_0 = 2\frac{\pi}{T_0}$ rad/s. The set of complex exponentials

$$\{\varphi_k(t) = e^{jk\omega_0 t}, \quad k \in \mathbb{Z}\} \tag{29}$$

are called **harmonically related** because each has a frequency $k\omega_0$ that is an integer multiple (harmonic) of ω_0. Each $\varphi_{k(t)}$ is itself periodic with period T_0.

3.2.1. Orthogonality

The harmonically related exponentials satisfy the orthogonality relation

$$\frac{1}{T_0}\int_0^{T_0} e^{jk\omega_0 t} e^{-jm\omega_0 t}\,\mathrm{d}t = \begin{cases} 1 \text{ if } k = m \\ 0 \text{ if } k \neq m. \end{cases} \tag{30}$$

Proof: For $k \neq m$, the integrand is $e^{j(k-m)\omega_0 t}$. Since $k - m$ is a nonzero integer, integrating over exactly one period T_0 gives zero. For $k = m$ the integrand is 1 and the integral equals $\frac{T_0}{T_0} = 1$.

This orthogonality plays the same role as the dot product of orthogonal basis vectors: it allows each Fourier coefficient to be extracted independently.

3.3. The Exponential Fourier Series

3.3.1. Synthesis equation

A periodic signal $x(t)$ with period T_0 can be represented as a linear combination of harmonically related complex exponentials:

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}. \tag{31}$$

This is the **exponential Fourier Series**. The complex constants c_n are the **Fourier coefficients**.

3.3.2. Analysis equation

Multiplying both sides of the synthesis equation by $e^{-jm\omega_0 t}$ and integrating over one period:

$$\begin{aligned} \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jm\omega_0 t}\,\mathrm{d}t &= \frac{1}{T_0}\int_0^{T_0} \sum_n c_n e^{jn\omega_0 t}e^{-jm\omega_0 t}\,\mathrm{d}t \\ &= \sum_n c_n \underbrace{\frac{1}{T_0}\int_0^{T_0} e^{j(n-m)\omega_0 t}\,\mathrm{d}t}_{=\delta[n-m]} \\ &= c_m. \end{aligned} \tag{32}$$

Therefore the Fourier coefficients are given by

$$c_n = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jn\omega_0 t}\,\mathrm{d}t. \tag{33}$$

The integration can be carried out over **any** interval of length T_0, not just $[0, T_0]$. The coefficient $c_0 = \left(\frac{1}{T_0}\right)\int_0^{T_0} x(t)\,\mathrm{d}t$ is the time-average (DC value) of $x(t)$.

3.4. The Trigonometric Fourier Series

For a **real-valued** signal $x(t)$, the complex coefficients satisfy the **conjugate symmetry** property $c_{-n} = c_n^*$. Grouping positive and negative frequency terms:

$$\begin{aligned} x(t) &= c_0 + \sum_{n=1}^{\infty}\left(c_n e^{jn\omega_0 t} + c_{-n}e^{-jn\omega_0 t}\right) \\ &= a_0 + \sum_{n=1}^{\infty}\left[a_n\cos(n\omega_0 t) + b_n\sin(n\omega_0 t)\right], \end{aligned} \tag{34}$$

where

$$a_0 = c_0, \quad a_n = 2\ \mathrm{Re}\{c_n\}, \quad b_n = -2\ \mathrm{Im}\{c_n\}. \tag{35}$$

Equivalently, combining cosine and sine into a single sinusoid:

$$x(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \varphi_n), \tag{36}$$

where $A_n = 2|c_n|$ and $\varphi_n = \langle c_n$. This is the **compact trigonometric** (or amplitude-phase) form.

> **Note**
>
> For real even signals $b_n = 0$ (cosine series only). For real odd signals $a_n = 0$ (sine series only). Recognising signal symmetry before computing coefficients can halve the integration work.

3.5. Convergence of the Fourier Series

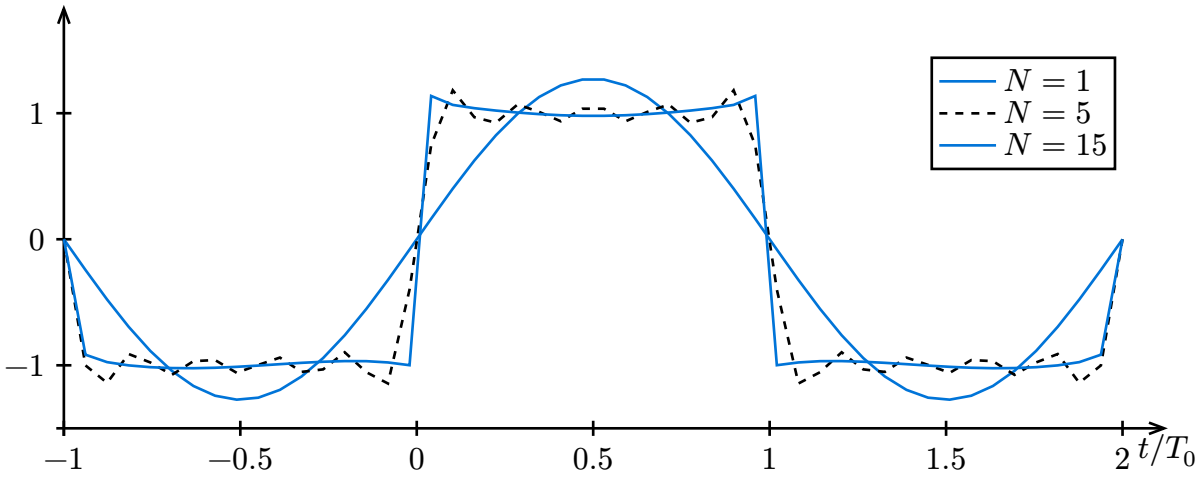
The Fourier Series does not converge for every periodic signal. A set of **sufficient** conditions (Dirichlet conditions) guarantees convergence:

1. $x(t)$ is absolutely integrable over one period: $\int_0^{T_0} |x(t)|\,\mathrm{d}t < \infty$.
2. $x(t)$ has a finite number of maxima and minima per period.
3. $x(t)$ has a finite number of discontinuities per period, each of finite size.

When these hold, the Fourier Series converges to $x(t)$ at every point of continuity, and to the average frac $\{1\}\{2\}[x(t^+) + x(t^-)]$ at each discontinuity.

3.5.1. Gibbs Phenomenon

At a jump discontinuity, the partial sum $x_{N(t)} = \sum_{n=-N}^{N} c_n e^{jn\omega_0 t}$ overshoots the signal value by approximately **9% of the jump magnitude**, regardless of how large N is. This overshoot does not diminish as $N \to \infty$; it only narrows. This is the **Gibbs phenomenon**.



3.6. Worked Examples

3.6.1. Square Wave

Let $x(t)$ be the square wave with amplitude A and period T_0:

$$x(t) = \begin{cases} A \text{ if } 0 \le t < \frac{T_0}{2} \\ -A \text{ if } \frac{T_0}{2} \le t < T_0. \end{cases} \tag{37}$$

Computing the Fourier coefficients using the analysis equation:

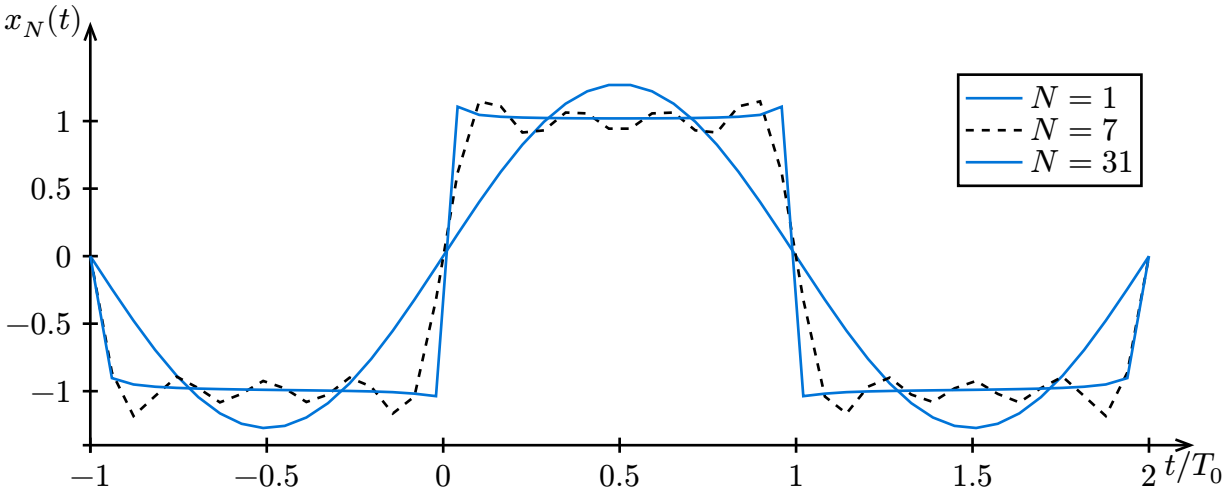
$$c_n = \frac{1}{T_0}\left[\int_0^{T_0/2} Ae^{-jn\omega_0 t}\,\mathrm{d}t + \int_{T_0/2}^{T_0} (-A)e^{-jn\omega_0 t}\,\mathrm{d}t\right]. \tag{38}$$

For $n \neq 0$, integrating and simplifying using $\omega_0 T_0 = 2\pi$:

$$c_n = \frac{A}{jn\pi}(1 - e^{-jn\pi}) = \begin{cases} \frac{2A}{jn\pi} n \text{ odd} \\ 0 \quad n \text{ even.} \end{cases} \tag{39}$$

For $n = 0$: $c_0 = 0$ (the square wave has zero mean). The Fourier Series contains only **odd harmonics**, and the coefficient magnitude decays as $\frac{1}{n}$:

$$x(t) = \frac{4A}{\pi} \sum_{k=0}^{\infty} \frac{1}{2k+1} \sin((2k+1)\omega_0 t). \tag{40}$$



3.6.2. Periodic Rectangular Pulse Train

Let $x(t)$ be a periodic train of rectangular pulses with amplitude A, pulse width τ, and period $T_0 > \tau$:

$$x(t) = A, \quad |t| \le \frac{\tau}{2} \text{ within each period.} \tag{41}$$

The Fourier coefficients are

$$c_n = \frac{1}{T_0}\int_{-\tau/2}^{\tau/2} Ae^{-jn\omega_0 t}\,\mathrm{d}t = \frac{A\tau}{T_0}\,\mathrm{sinc}\left(n\omega_0\frac{\tau}{2}\right), \tag{42}$$

where $\mathrm{sinc}(u) = \frac{\sin(u)}{u}$ (unnormalised). Written in terms of the duty cycle $D = \frac{\tau}{T_0}$:

$$c_n = AD, \mathrm{sinc}(n\pi D). \tag{43}$$

Key observations:

- $c_0 = AD$: the DC value equals amplitude times duty cycle.
- Spectral nulls occur at $n = \frac{1}{D}, \frac{2}{D}, \ldots$ (whenever $n\pi D$ is a multiple of π).
- Narrower pulses ($D \to 0$) produce a **wider** spectral envelope and more visible harmonics — an instance of the **time–bandwidth trade-off**.

3.6.3. Full-Wave Rectified Sinusoid

Let $x(t) = A|\cos(\omega_0 t)|$. Because the rectification doubles the frequency, $x(t)$ has fundamental period $\frac{T_0}{2}$ and angular frequency $2\omega_0$. The Fourier coefficients evaluate to

$$c_n = \frac{2A}{\pi}\frac{(-1)^n}{1-4n^2}. \tag{44}$$

Only **even** harmonics of the original ω_0 appear (n = 0, ±2, ±4, ...), contrasting with the square wave which contains only odd harmonics. The coefficients decay as $\frac{1}{n^2}$ — faster than the square wave's $\frac{1}{n}$ — because the rectified sinusoid is continuous (no jump discontinuities).

3.7. Properties of the Continuous-Time Fourier Series

Let $x(t) \overset{\text{FS}}{\longleftrightarrow} c_n$ and $y(t) \overset{\text{FS}}{\longleftrightarrow} d_n$, both with period T_0 and $\omega_0 = 2\frac{\pi}{T_0}$.

Property	Time domain	Fourier coefficients
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha c_n + \beta d_n$
Time shifting	$x(t - t_0)$	$c_n e^{-jn\omega_0 t_0}$
Time reversal	$x(-t)$	c_{-n}
Conjugation	$x^*(t)$	c^*_{-n}
Conj. symmetry	$x(t)$ real	$c_{-n} = c_n^*$
Multiplication	$x(t) \cdot y(t)$	$\sum_k c_k d_{n-k}$
Parseval's thm.	$(1/T_0)\int_0^{T_0} \lvert x(t)\rvert^2\,\mathrm{d}t$	$\sum_{n=-\infty}^{\infty} \lvert c_n\rvert^2$

3.7.1. Proofs of selected properties

Time shifting: Replace t with $t - t_0$ in the synthesis equation:

$$x(t-t_0) = \sum_n c_n e^{jn\omega_0(t-t_0)} = \sum_n (c_n e^{-jn\omega_0 t_0}) e^{jn\omega_0 t}. \tag{45}$$

The coefficient of $e^{jn\omega_0 t}$ is $c_n e^{-jn\omega_0 t_0}$. The **magnitude** $|c_n|$ is unchanged; only the **phase** shifts by $-n\omega_0 t_0$.

Parseval's theorem: The average power of $x(t)$ equals the sum of powers in each harmonic:

$$\frac{1}{T_0}\int_0^{T_0} |x(t)|^2\,\mathrm{d}t = \sum_{n=-\infty}^{\infty} |c_n|^2. \tag{46}$$

Proof: substitute the synthesis equation for one factor of $x(t)$ and use orthogonality.

3.8. The Frequency Spectrum

The **amplitude spectrum** $|c_n|$ and **phase spectrum** $\angle c_n$ plotted against the harmonic index n (or frequency $n\omega_0$) form the **line spectrum** of $x(t)$. Because n takes only integer values, the spectrum is **discrete** — energy exists only at $\omega = 0, \pm\omega_0, \pm 2\omega_0, \ldots$

Key observations:

- For real $x(t)$: $|c_{-n}| = |c_n|$ (amplitude spectrum is even) and $\angle c_{-n} = -\angle c_n$ (phase spectrum is odd).
- The spectral **envelope** shape is determined by the signal's time-domain shape; for the rect pulse train it follows a $|\text{sinc}|$ curve.
- Decreasing the period T_0 (higher repetition rate) **increases** the harmonic spacing $\omega_0 = 2\frac{\pi}{T_0}$.
- Decreasing the pulse width τ (for fixed T_0) **widens** the sinc envelope, spreading energy to higher harmonics.

3.9. Fourier Series Analysis of LTI Systems

From §2.1, the response of an LTI system with frequency response $H(j\omega)$ to $e^{jn\omega_0 t}$ is $H(jn\omega_0)e^{jn\omega_0 t}$. By linearity, for a periodic input $x(t) = \sum_n c_n e^{jn\omega_0 t}$ the steady-state output is

$$y(t) = \sum_{n=-\infty}^{\infty} c_n H(jn\omega_0) e^{jn\omega_0 t}. \tag{47}$$

The output Fourier coefficients are $d_n = c_n H(jn\omega_0)$: each harmonic is scaled in amplitude by $|H(jn\omega_0)|$ and shifted in phase by $\angle H(jn\omega_0)$.

3.9.1. Worked example: RC lowpass filter driven by a square wave

Consider a first-order RC lowpass filter with frequency response

$$H(j\omega) = \frac{1}{1 + j\omega RC}. \tag{48}$$

The filter has $|H(j\omega)| = \frac{1}{\sqrt{1+(\omega RC)^2}}$ — it attenuates high-frequency harmonics. Let the input be the square wave from §2.6.1 with $A = 1$, $T_0 = 2\pi$ (so $\omega_0 = 1$ rad/s), and let $RC = 1$ s. The input Fourier coefficients are $c_n = \frac{2}{jn\pi}$ for odd n, 0 otherwise. The output coefficients are

$$d_n = c_n H(jn) = \frac{2}{jn\pi} \cdot \frac{1}{1 + jn}, \quad n \text{ odd}. \tag{49}$$

The output is dominated by the fundamental ($n = \pm 1$) since higher harmonics are progressively attenuated. As RC increases the filter becomes more selective and the output approaches a sinusoid.

4. Continuous-Time Fourier Transform

4.1. From Fourier Series to the Fourier Transform

In Chapter 2 we represented **periodic** signals as a discrete sum of complex exponentials. Most signals of engineering interest — a single pulse, a decaying transient, a finite-duration waveform — are **aperiodic**. To handle these we extend the Fourier Series by considering what happens when the period T_0 grows without bound.

4.1.1. The Limiting Argument

Consider a periodic signal $\tilde{x}(t)$ with period T_0 whose single period $|t| < \frac{T_0}{2}$ matches a given aperiodic signal $x(t)$. As $T_0 \to \infty$ the periodic replica moves infinitely far from the origin and $\tilde{x}(t) \to x(t)$ everywhere.

The Fourier coefficients of $\tilde{x}(t)$ are

$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \tilde{x}(t) e^{-jn\omega_0 t}\, \mathrm{d}t. \tag{50}$$

Define the **spectral envelope** $X(j\omega)$ by

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, \mathrm{d}t, \tag{51}$$

so that $T_0 c_n = X(jn\omega_0)$, i.e. $c_n = \frac{X(jn\omega_0)}{T_0}$.

Substituting into the synthesis equation:

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{\infty} \frac{X(jn\omega_0)}{T_0} e^{jn\omega_0 t} = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} X(jn\omega_0) e^{jn\omega_0 t} \Delta\omega, \tag{52}$$

where $\Delta\omega = \omega_0 = 2\frac{\pi}{T_0}$. As $T_0 \to \infty$, the harmonic spacing $\Delta\omega \to \mathrm{d}\omega$ and the discrete sum passes to a Riemann integral:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\, \mathrm{d}\omega. \tag{53}$$

This is the **inverse Fourier transform** (synthesis equation). The companion **forward transform** (analysis equation) is identified as

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, \mathrm{d}\omega. \tag{54}$$

Key insight

The passage from Fourier Series to Fourier Transform is the passage from a discrete spectrum (line spectrum at harmonics $n\omega_0$) to a **continuous spectrum** $X(j\omega)$. The Fourier coefficient c_n is replaced by the spectral density $X(j\omega)/2\pi$ — a density because energy is now spread continuously over frequency rather than concentrated at discrete lines.

4.1.2. The Spectrum

$X(j\omega)$ is in general complex. Writing it in polar form

$$X(j\omega) = |X(j\omega)|\ e^{j\langle X(j\omega)}, \tag{55}$$

we call $|X(j\omega)|$ the **magnitude spectrum** and $\langle X(j\omega)$ the **phase spectrum** of $x(t)$. Together they carry all the frequency-domain information about the signal.

4.2. Definition and Existence

4.2.1. The Fourier Transform Pair

The **Continuous-Time Fourier Transform (CTFT)** of a signal $x(t)$ and its inverse are defined by the pair

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,\mathrm{d}t \qquad \text{(analysis / forward transform)} \tag{56}$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\,\mathrm{d}\omega \quad \text{(synthesis / inverse transform)} \tag{57}$$

We write this relationship compactly as $x(t) \overset{\mathcal{F}}{\leftrightarrows} X(j\omega)$, or equivalently $X(j\omega) = \mathcal{F}\{x(t)\}$ and $x(t) = \mathcal{F}^{-1}\{X(j\omega)\}$.

Note

Two sign conventions exist in the literature. This course follows the **engineering convention**: $e^{-j\omega t}$ in the forward transform and $e^{+j\omega t}$ in the inverse, with the $1/(2\pi)$ factor on the inverse. Oppenheim & Willsky and Lathi both use this convention.

4.2.2. Conditions for Existence (Dirichlet Conditions)

Not every signal has a Fourier transform in the classical sense. The following **Dirichlet conditions** are **sufficient** (not necessary) to guarantee that $X(j\omega)$ exists and that the inverse transform reconstructs $x(t)$:

1. **Absolute integrability:** $\int_{-\infty}^{\infty} |x(t)|\,\mathrm{d}t < \infty$.
2. **Finite extrema:** $x(t)$ has a finite number of maxima and minima in every finite interval.
3. **Finite discontinuities:** $x(t)$ has a finite number of jump discontinuities in every finite interval, each of finite size.

When these conditions hold, $X(j\omega)$ is well-defined and the inverse transform converges to $x(t)$ at every point of continuity, and to the average $\text{frac}\{1\}\{2\}[x(t^+) + x(t^-)]$ at each jump.

Note

Several signals that violate condition 1 — notably $u(t)$, $\cos(\omega_0 t)$, and the unit impulse $\delta(t)$ — are so important in system analysis that their transforms are defined using **generalised functions** (distributions). These involve Dirac deltas in the frequency domain and are covered in §4.3 and §4.4.

4.3. Common Fourier Transform Pairs

This section derives the transforms of the signals most frequently encountered in system analysis. Each result is worth memorising; a summary table is given at the end.

4.3.1. Rectangular Pulse (Gate Function)

Define the rectangular pulse of width τ and unit amplitude:

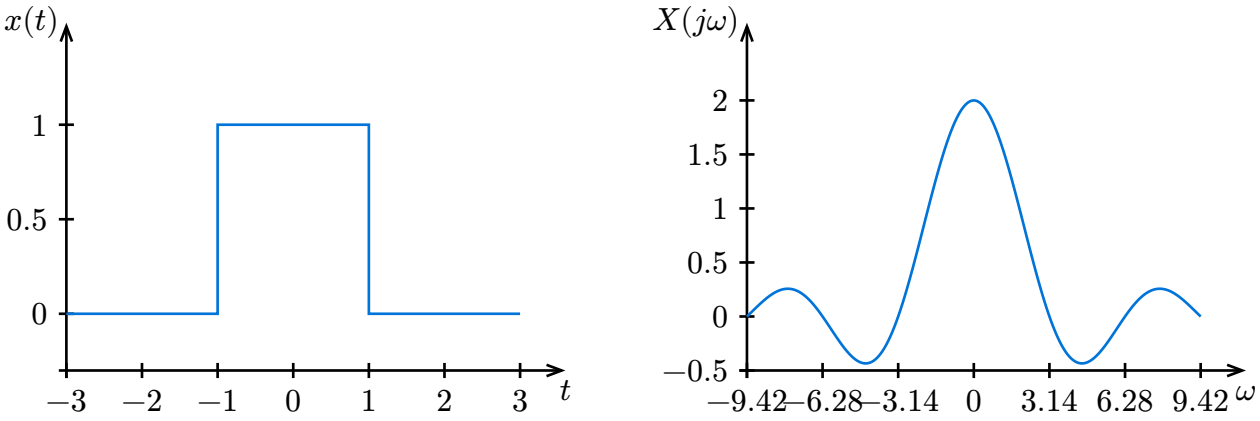
$$x(t) = \operatorname{rect}(t/\tau) = \begin{cases} 1 & |t| \le \tau/2 \\ 0 & |t| > \tau/2. \end{cases} \tag{58}$$

Applying the forward transform:

$$\begin{aligned} X(j\omega) &= \int_{-\tau/2}^{\tau/2} e^{-j\omega t}\,\mathrm{d}t = \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-\tau/2}^{\tau/2} = \frac{e^{-j\omega\tau/2} - e^{j\omega\tau/2}}{-j\omega} \\ &= \frac{2\sin(\omega\tau/2)}{\omega} = \tau \cdot \frac{\sin(\omega\tau/2)}{\omega\tau/2} = \tau\,\operatorname{sinc}(\omega\tau/2), \end{aligned} \tag{59}$$

where $\operatorname{sinc}(u) = \sin(u)/u$ (unnormalised). Therefore

$$\operatorname{rect}(t/\tau) \overset{\mathcal{F}}{\leftrightarrow} \tau\,\operatorname{sinc}(\omega\tau/2). \tag{60}$$



4.3.2. Decaying Exponential

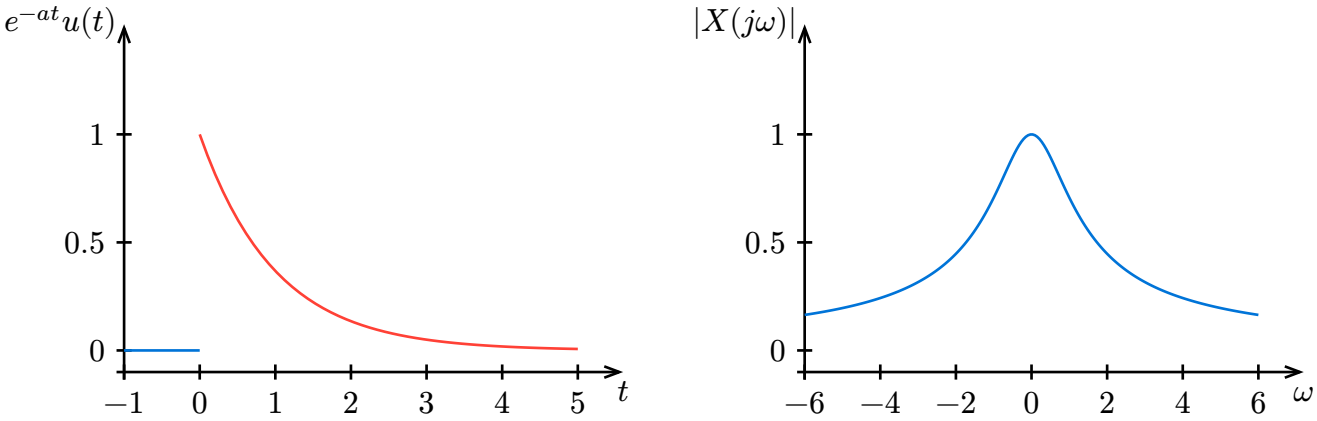
For $a > 0$, consider the causal decaying exponential $x(t) = e^{-at}u(t)$:

$$X(j\omega) = \int_0^\infty e^{-at}e^{-j\omega t}\,\mathrm{d}t = \int_0^\infty e^{-(a+j\omega)t}\,\mathrm{d}t = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)}\right]_0^\infty = \frac{1}{a+j\omega}. \quad (61)$$

The integral converges because $a > 0$ ensures $e^{-(a+j\omega)t} \to 0$ as $t \to \infty$. Therefore

$$e^{-at}u(t) \overset{\mathcal{F}}{\leftrightarrows} \frac{1}{a+j\omega}, \quad a > 0. \quad (62)$$

The magnitude spectrum is a **Lorentzian**: $|X(j\omega)| = 1/\sqrt{a^2+\omega^2}$, which decays as $\frac{1}{\omega}$ for large $|\omega|$. The 3 dB bandwidth (where $|X|$ falls to $1/\sqrt{2}$ of its DC value) is $\omega_{3\text{dB}} = a$ rad/s.



4.3.3. Two-Sided Decaying Exponential

For $a > 0$, let $x(t) = e^{-a|t|}$. Split the integral at $t = 0$:

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^0 e^{at}e^{-j\omega t}\,\mathrm{d}t + \int_0^\infty e^{-at}e^{-j\omega t}\,\mathrm{d}t \\ &= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} = \frac{2a}{a^2+\omega^2}. \end{aligned} \quad (63)$$

This is a real, even, positive function of ω — the phase spectrum is zero everywhere.

4.3.4. Unit Impulse

$$\mathcal{F}\{\delta(t)\} = \int_{-\infty}^\infty \delta(t)e^{-j\omega t}\,\mathrm{d}t = e^{-j\omega\cdot 0} = 1. \quad (64)$$

The impulse has a **flat** (white) spectrum: all frequencies are present with equal magnitude. For a time-shifted impulse:

$$\mathcal{F}\{\delta(t-t_0)\} = e^{-j\omega t_0}. \quad (65)$$

The magnitude is still 1 for all ω; only the phase $-\omega t_0$ changes.

4.3.5. Constant Signal and the DC Impulse

Taking the inverse transform of $2\pi\delta(\omega)$:

$$\frac{1}{2\pi}\int_{-\infty}^\infty 2\pi\delta(\omega)e^{j\omega t}\,\mathrm{d}\omega = 1. \quad (66)$$

Therefore $\mathcal{F}\{1\} = 2\pi\delta(\omega)$. This result makes physical sense: a constant signal is pure DC, so its entire energy is concentrated at $\omega = 0$.

4.3.6. Signum Function

The signum function $\operatorname{sgn}(t) = u(t) - u(-t)$ is an odd signal. It can be obtained as a limit of the two-sided exponential:

$$\operatorname{sgn}(t) = \lim_{a \to 0} \begin{cases} e^{-at} & t > 0 \\ -e^{at} & t < 0. \end{cases} \tag{67}$$

Taking the transform and letting $a \to 0$:

$$\mathcal{F}\{\operatorname{sgn}(t)\} = \lim_{a \to 0} \left(\frac{1}{a + j\omega} - \frac{1}{a - j\omega} \right) = \frac{2}{j\omega}. \tag{68}$$

4.3.7. Unit Step

Since $u(t) = \frac{1}{2} + \frac{1}{2}\operatorname{sgn}(t)$, linearity gives

$$\mathcal{F}\{u(t)\} = \pi\delta(\omega) + \frac{1}{j\omega}. \tag{69}$$

The $\pi\delta(\omega)$ term captures the nonzero mean (DC component); the $1/(j\omega)$ term accounts for the step edge.

4.3.8. Summary Table of Transform Pairs

Signal $x(t)$	Transform $X(j\omega)$
$\operatorname{rect}(t/\tau)$	$\tau, \operatorname{sinc}(\omega\tau/2)$
$\operatorname{sinc}(Wt/2)/\pi$	$\operatorname{rect}(\omega/(2W))$
$e^{-at}u(t), a > 0$	$1/(a + j\omega)$
$te^{-at}u(t), a > 0$	$1/(a + j\omega)^2$
$e^{-a\lvert t\rvert}, a > 0$	$2a/(a^2 + \omega^2)$
e^{-at^2}	$\sqrt{\pi/a}e^{-\omega^2/(4a)}$
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
1	$2\pi\delta(\omega)$
$\operatorname{sgn}(t)$	$2/(j\omega)$
$u(t)$	$\pi\delta(\omega) + 1/(j\omega)$
$\cos(\omega_0 t)$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

4.4. Fourier Transform of Periodic Signals

A periodic signal $x(t)$ with period T_0 violates the absolute integrability condition, so its Fourier transform does not exist in the classical sense. However, we can use the Fourier Series representation together with the result $\mathcal{F}\{1\} = 2\pi\delta(\omega)$ to assign a transform that is consistent with the inverse transform formula.

4.4.1. Derivation

Write the Fourier Series of $x(t)$:

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}. \tag{70}$$

Taking the Fourier transform term-by-term and using $\mathcal{F}\{e^{jn\omega_0 t}\} = 2\pi\delta(\omega - n\omega_0)$:

$$X(j\omega) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0). \tag{71}$$

The spectrum of a periodic signal is a **train of impulses** in frequency, located at the harmonics $\omega = n\omega_0$, each weighted by $2\pi c_n$.

Key insight

This result unifies periodic and aperiodic signals under one framework: both have a Fourier transform, but periodic signals have a **discrete** (impulsive) spectrum while aperiodic signals have a **continuous** spectrum. The Fourier Series is simply the special case of the Fourier Transform for periodic signals.

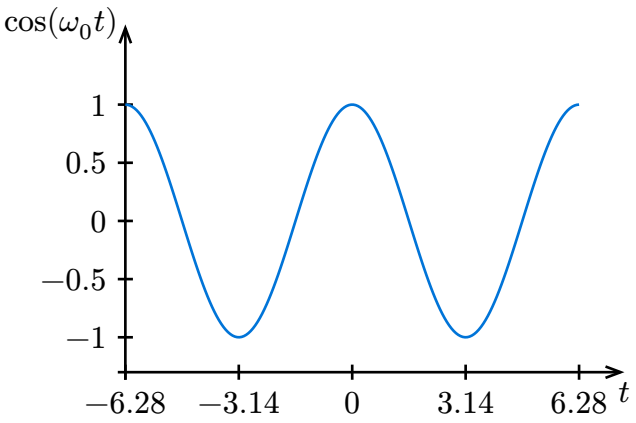
4.4.2. Example: Sinusoid

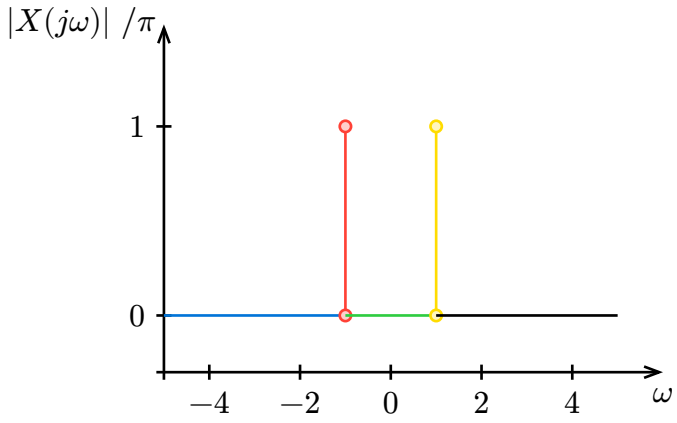
For $x(t) = \cos(\omega_0 t) = \text{frac}\{1\}\{2\}(e^{j\omega_0 t} + e^{-j\omega_0 t})$, the Fourier Series has $c_1 = c_{-1} = \frac{1}{2}$ and all other $c_n = 0$. Therefore

$$\mathcal{F}\{\cos(\omega_0 t)\} = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]. \tag{72}$$

Two impulses of weight π sit symmetrically at $\pm\omega_0$. Similarly

$$\mathcal{F}\{\sin(\omega_0 t)\} = j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]. \tag{73}$$





4.4.3. Example: Periodic Impulse Train

The **periodic impulse train** (Shah function) is defined as

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0). \tag{74}$$

Its Fourier Series coefficients are $c_n = \frac{1}{T_0}$ for all n (since each impulse is identical). Therefore

$$\mathcal{F}\{p(t)\} = \frac{2\pi}{T_0} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0) = \omega_0 \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0). \tag{75}$$

A periodic impulse train in time transforms to a periodic impulse train in frequency, with spacing $\omega_0 = 2\pi/T_0$. This result is fundamental to the derivation of the Nyquist Sampling Theorem (Chapter 6).

> **Note**
>
> The impulse train result shows the **inverse relationship** between time and frequency: a denser train in time (T_0 small, ω_0 large) produces a sparser train in frequency, and vice versa. This is a direct manifestation of the time–frequency uncertainty principle.

4.5. Properties of the Fourier Transform

The properties below allow transforms of complicated signals to be deduced from simpler known pairs, and are indispensable for solving system analysis problems. Throughout, assume $x(t) \overset{\mathcal{F}}{\leftrightarrow} X(j\omega)$ and $y(t) \overset{\mathcal{F}}{\leftrightarrow} Y(j\omega)$.

4.5.1. Linearity

For constants $a, b \in \mathbb{C}$:

$$ax(t) + by(t) \overset{\mathcal{F}}{\leftrightarrow} aX(j\omega) + bY(j\omega). \tag{76}$$

Proof: The Fourier integral is a linear operation — constants factor out and integrals of sums split. □

4.5.2. Time Shifting

$$x(t - t_0) \overset{\mathcal{F}}{\leftrightarrow} e^{-j\omega t_0} X(j\omega). \tag{77}$$

Proof: Substitute $\tau = t - t_0$ in the forward transform:

$$\int_{-\infty}^{\infty} x(t - t_0) e^{-j\omega t}\, \mathrm{d}t = \int_{-\infty}^{\infty} x(\tau) e^{-j\omega(\tau + t_0)}\, \mathrm{d}\tau = e^{-j\omega t_0} X(j\omega). \quad \square \tag{78}$$

Time shifting leaves the **magnitude spectrum unchanged** ($|e^{-j\omega t_0} X(j\omega)| = |X(j\omega)|$) and adds a **linear phase** $-\omega t_0$ to the phase spectrum. A delay in time corresponds to a linear phase ramp in frequency.

4.5.3. Frequency Shifting (Modulation)

$$e^{j\omega_0 t}x(t) \overset{\mathcal{F}}{\leftrightarrows} X(j(\omega-\omega_0)). \tag{79}$$

Proof: Replace ω with $\omega-\omega_0$ in the analysis integral:

$$\int_{-\infty}^{\infty} e^{j\omega_0 t}x(t)e^{-j\omega t}\,\mathrm{d}t = \int_{-\infty}^{\infty} x(t)e^{-j(\omega-\omega_0)t}\,\mathrm{d}t = X(j(\omega-\omega_0)). \quad \square \tag{80}$$

Multiplication by a complex exponential **shifts the spectrum** by ω_0. This is the mathematical basis of **amplitude modulation** in communications.

Example: $x(t)\cos(\omega_0 t) = \tfrac{1}{2}x(t)(e^{j\omega_0 t}+e^{-j\omega_0 t})$, so

$$\mathcal{F}\{x(t)\cos(\omega_0 t)\} = \tfrac{1}{2}[X(j(\omega-\omega_0)) + X(j(\omega+\omega_0))]. \tag{81}$$

The spectrum of $x(t)$ is translated to be centred at $\pm\omega_0$.

4.5.4. Time and Frequency Scaling

For real $a \neq 0$:

$$x(at) \overset{\mathcal{F}}{\leftrightarrows} \frac{1}{|a|}X\!\left(\frac{j\omega}{a}\right). \tag{82}$$

Proof: For $a>0$, substitute $\tau = at$:

$$\int_{-\infty}^{\infty} x(at)e^{-j\omega t}\,\mathrm{d}t = \frac{1}{a}\int_{-\infty}^{\infty} x(\tau)e^{-j(\omega/a)\tau}\,\mathrm{d}\tau = \frac{1}{a}X\!\left(\frac{j\omega}{a}\right). \tag{83}$$

For $a<0$ the limits of integration reverse, introducing a factor $1/|a|$. $\square$

Compressing $x(t)$ in time ($|a|>1$) expands its spectrum in frequency and reduces its amplitude, and vice versa. This is the **time–bandwidth trade-off**: you cannot simultaneously make a signal narrow in time and narrow in frequency.

4.5.5. Time Reversal

Setting $a=-1$ in the scaling property:

$$x(-t) \overset{\mathcal{F}}{\leftrightarrows} X(-j\omega) = X^{*}(j\omega), \quad x(t)\ \text{real}. \tag{84}$$

Reversing a real signal conjugates its spectrum, flipping the sign of the phase while leaving the magnitude unchanged.

4.5.6. Duality

If $x(t) \overset{\mathcal{F}}{\leftrightarrows} X(j\omega)$, then

$$X(jt) \overset{\mathcal{F}}{\leftrightarrows} 2\pi x(-\omega). \tag{85}$$

Proof: Starting from the inverse transform $x(t) = \frac{1}{2\pi}\int X(j\omega)e^{j\omega t}\,\mathrm{d}\omega$, replace t with $-t$ and then swap the roles of t and ω. $\square$

Duality means every transform pair generates a second pair for free.

Example: Since $\operatorname{rect}(t/\tau) \overset{\mathcal{F}}{\leftrightarrows} \tau, \operatorname{sinc}(\omega\tau/2)$, duality gives

$$\tau, \operatorname{sinc}(t\tau/2) \overset{\mathcal{F}}{\leftrightarrows} 2\pi, \operatorname{rect}(-\omega/\tau) = 2\pi, \operatorname{rect}(\omega/\tau). \tag{86}$$

A sinc pulse in time has a rectangular (bandlimited) spectrum — it is the ideal low-pass filter impulse response.

4.5.7. Differentiation in Time

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} \overset{\mathcal{F}}{\leftrightarrows} j\omega X(j\omega). \tag{87}$$

Proof: Differentiate the inverse transform formula under the integral sign:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\text{frac}\{1\}\{2\pi\}\int X(j\omega)e^{j\omega t}\,\mathrm{d}\omega\right] = \text{frac}\{1\}\{2\pi\}\int j\omega X(j\omega)e^{j\omega t}\,\mathrm{d}\omega. \quad \square \tag{88}$$

By induction, differentiation n times gives $(j\omega)^n X(j\omega)$. This property converts a differential equation in t into an **algebraic equation** in ω — the central tool of Fourier-domain system analysis.

4.5.8. Integration in Time

$$\int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau \overset{\mathcal{F}}{\leftrightarrows} \text{frac}\{X(j\omega)\}\{j\omega\} + \pi X(0)\delta(\omega). \tag{89}$$

The $\pi X(0)\delta(\omega)$ term accounts for any nonzero area (DC content) accumulated by the integral as $t \to \infty$.

4.5.9. Convolution in Time

$$x(t) * y(t) \overset{\mathcal{F}}{\leftrightarrows} X(j\omega)Y(j\omega). \tag{90}$$

Proof: Write the convolution and forward transform, then exchange the order of integration:

$$\begin{aligned}
\mathcal{F}\{x*y\} &= \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} x(\tau)y(t-\tau)\,\mathrm{d}\tau\right]e^{-j\omega t}\,\mathrm{d}t \\
&= \int_{-\infty}^{\infty} x(\tau)\left[\int_{-\infty}^{\infty} y(t-\tau)e^{-j\omega t}\,\mathrm{d}t\right]\mathrm{d}\tau \\
&= \int_{-\infty}^{\infty} x(\tau)e^{-j\omega\tau}Y(j\omega)\,\mathrm{d}\tau \\
&= X(j\omega)Y(j\omega). \quad \square
\end{aligned} \tag{91}$$

Key insight

The convolution theorem is the most important property in system analysis. Convolution in time — a laborious integral — becomes **pointwise multiplication** in frequency. To find a system's output: transform the input, multiply by $H(j\omega)$, inverse transform.

4.5.10. Multiplication in Time (Windowing)

$$x(t) \cdot y(t) \overset{\mathcal{F}}{\leftrightarrows} \text{frac}\{1\}\{2\pi\} X(j\omega) * Y(j\omega). \tag{92}$$

Multiplication in time corresponds to convolution in frequency (with a $1/2\pi$ normalisation). This property underlies **windowing** in spectral analysis: multiplying a signal by a window function $w(t)$ convolves its spectrum with $W(j\omega)$, which must be chosen narrow in frequency to minimise spectral leakage.

4.5.11. Parseval's Theorem

$$\int_{-\infty}^{\infty} |x(t)|^2 \, \mathrm{d}t = \text{frac}\{1\}\{2\pi\} \int_{-\infty}^{\infty} |X(j\omega)|^2 \, \mathrm{d}\omega. \tag{93}$$

Proof: Apply the multiplication property with $y(t) = x^*(t)$, noting that $\mathcal{F}\{x^*(t)\} = X^*(-j\omega)$, then evaluate at $\omega = 0$. □

Parseval's theorem states that **total signal energy is preserved** by the Fourier transform. The quantity $|X(j\omega)|^2/(2\pi)$ is the **energy spectral density** — it describes how the signal's energy is distributed across frequency.

4.5.12. Summary Table of Properties

Property	Time domain	Frequency domain
Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
Time shifting	$x(t - t_0)$	$e^{-j\omega t_0} X(j\omega)$
Freq. shifting	$e^{j\omega_0 t} x(t)$	$X(j(\omega - \omega_0))$
Time scaling	$x(at)$	$\text{frac}\{1\}\{\lvert a\rvert\} X(j\omega/a)$
Time reversal	$x(-t)$	$X(-j\omega)$
Duality	$X(jt)$	$2\pi x(-\omega)$
Differentiation	$\mathrm{d}^n x / \mathrm{d}t^n$	$(j\omega)^n X(j\omega)$
Integration	$\int_{-\infty}^{t} x \, \mathrm{d}\tau$	$X(j\omega)/(j\omega) + \pi X(0)\delta(\omega)$
Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
Multiplication	$x(t)y(t)$	$\text{frac}\{1\}\{2\pi\} X(j\omega) * Y(j\omega)$
Parseval's thm.	$\int \lvert x\rvert^2 \, \mathrm{d}t$	$\text{frac}\{1\}\{2\pi\} \int \lvert X\rvert^2 \, \mathrm{d}\omega$

Property	Time domain	Frequency domain
Conj. symmetry	$x(t)$ real	$X(-j\omega) = X^*(j\omega)$

4.6. Frequency Response of LTI Systems

4.6.1. The Frequency Response $H(j\omega)$

From Chapter 1, the output of an LTI system with impulse response $h(t)$ is the convolution $y(t) = x(t) * h(t)$. Applying the convolution theorem:

$$Y(j\omega) = H(j\omega)X(j\omega), \tag{94}$$

where

$$H(j\omega) = \int_{-\infty}^{\infty} h(t)e^{-j\omega t}\,\mathrm{d}t = \mathcal{F}\{h(t)\}. \tag{95}$$

$H(j\omega)$ is the **frequency response** of the system. It is complex-valued and decomposes as

$$H(j\omega) = |H(j\omega)|\ e^{j\langle H(j\omega)}, \tag{96}$$

where $|H(j\omega)|$ is the **gain** (amplitude response) and $\langle H(j\omega)$ is the **phase response** at frequency ω. At each frequency ω, the system scales the input sinusoidal component by $|H(j\omega)|$ and shifts its phase by $\langle H(j\omega)$.

Key insight

The frequency response $H(j\omega)$ is the eigenvalue of the LTI system at frequency ω (recall §3.1). System analysis in the frequency domain reduces to: multiply spectra. The convolution integral, which is difficult to compute, is replaced by pointwise multiplication.

4.6.2. Ideal Filters

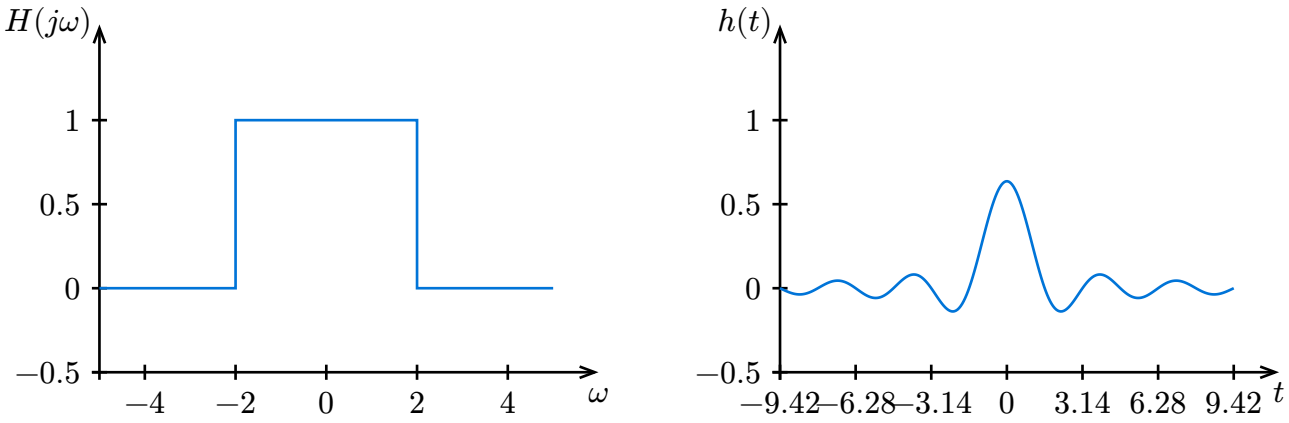
An **ideal filter** passes certain frequencies without distortion and completely blocks all others. The four ideal filter types are characterised by their frequency responses:

Filter type	Frequency response $H(j\omega)$
Low-pass (LPF)	$\mathrm{rect}(\omega/(2\omega_c))$
High-pass (HPF)	$1 - \mathrm{rect}(\omega/(2\omega_c))$
Band-pass (BPF)	$\mathrm{rect}((\omega-\omega_c)/B) + \mathrm{rect}((\omega+\omega_c)/B)$
Band-stop (BSF)	$1 - [\mathrm{rect}((\omega-\omega_c)/B) + \mathrm{rect}((\omega+\omega_c)/B)]$

where ω_c is the **cutoff frequency** and B is the **bandwidth**.

The impulse response of the ideal LPF is found by taking the inverse Fourier transform of $H(j\omega) = \mathrm{rect}(\omega/(2\omega_c))$:

$$h(t) = \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} e^{j\omega t}\,\mathrm{d}\omega = \frac{\omega_c}{\pi}\,\operatorname{sinc}(\omega_c t). \tag{97}$$



Note

The ideal LPF impulse response $h(t) = (\omega_c/\pi)\,\operatorname{sinc}(\omega_c t)$ is nonzero for $t < 0$ — it responds **before** the impulse is applied. This means the ideal LPF is **non-causal** and therefore physically unrealisable. Practical filter design (Chapter 5) constructs causal approximations with a smooth roll-off.

4.6.3. LTI Systems Described by Differential Equations

A broad class of physical systems — electrical circuits, mechanical systems, control loops — is described by linear constant-coefficient ordinary differential equations (LCCODEs) of the form

$$\sum_{k=0}^{N} a_k \frac{\mathrm{d}^k y}{\mathrm{d}t^k} = \sum_{k=0}^{M} b_k \frac{\mathrm{d}^k x}{\mathrm{d}t^k}. \tag{98}$$

Taking the Fourier transform of both sides and applying the differentiation property $\left(\mathrm{d}^k/\,\mathrm{d}t^k\right) \overset{\mathcal{F}}{\leftrightarrow} (j\omega)^k$:

$$\left(\sum_{k=0}^{N} a_k (j\omega)^k\right) Y(j\omega) = \left(\sum_{k=0}^{M} b_k (j\omega)^k\right) X(j\omega). \tag{99}$$

The frequency response is therefore a **rational function** of $j\omega$:

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k} = \frac{N(j\omega)}{D(j\omega)}. \tag{100}$$

4.6.3.1. Worked Example: Series RC Circuit

Consider a series RC circuit with input voltage $x(t)$ applied across the series combination, and output $y(t)$ taken across the capacitor. Kirchhoff's voltage law gives

$$RC\frac{\mathrm{d}y(t)}{\mathrm{d}t} + y(t) = x(t). \tag{101}$$

Taking the Fourier transform:

$$j\omega RCY(j\omega) + Y(j\omega) = X(j\omega) \quad \Rightarrow \quad H(j\omega) = \frac{1}{1 + j\omega RC}. \tag{102}$$

Let $\tau = RC$ (time constant). The magnitude and phase responses are

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}}, \quad \angle H(j\omega) = -\arctan(\omega\tau). \tag{103}$$

The gain is 1 at DC, falls to $1/\sqrt{2}$ (-3 dB) at $\omega_{3\text{dB}} = 1/\tau$, and rolls off as $1/\omega$ for $\omega \gg 1/\tau$. The phase shifts from 0 at DC to $-\pi/2$ at high frequency.

Now find the output when the input is $x(t) = e^{-at}u(t)$ with $a \neq 1/\tau$:

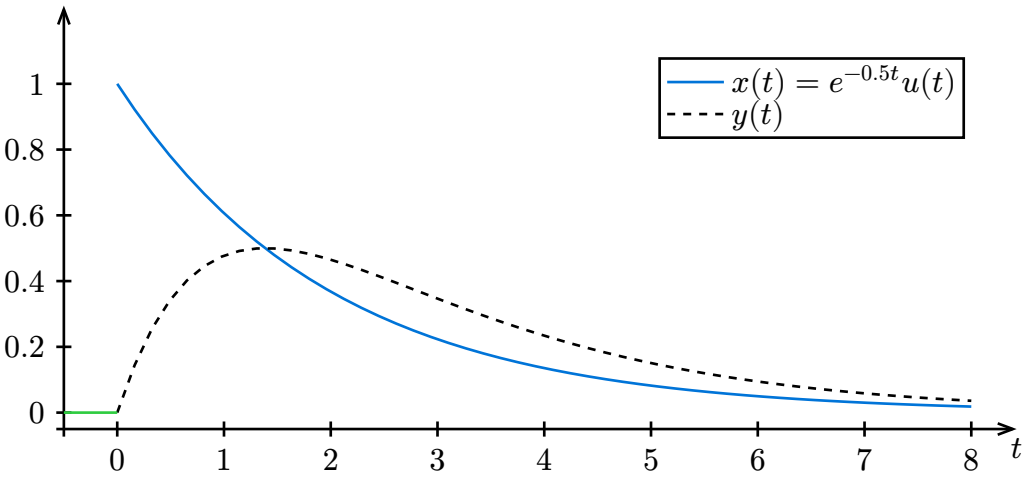
$$Y(j\omega) = H(j\omega)X(j\omega) = \frac{1}{1 + j\omega\tau} \cdot \frac{1}{a + j\omega}. \tag{104}$$

Partial fractions (with $b = 1/\tau$):

$$Y(j\omega) = \frac{1}{(a + j\omega)(b + j\omega)} = \frac{1}{b - a}\left(\frac{1}{a + j\omega} - \frac{1}{b + j\omega}\right). \tag{105}$$

Inverse transforming using $1/(\alpha + j\omega) \overset{\mathcal{F}^{-1}}{\leftrightarrow} e^{-\alpha t}u(t)$:

$$y(t) = \frac{1}{b - a}\left(e^{-at} - e^{-bt}\right)u(t) = \frac{\tau}{1 - a\tau}\left(e^{-at} - e^{-t/\tau}\right)u(t). \tag{106}$$



4.7. Signal Transmission and Distortion

When a signal passes through an LTI system the output is $Y(j\omega) = H(j\omega)X(j\omega)$. In general, the system modifies both the amplitude and phase of each frequency component. **Distortion** is any unwanted modification — the goal of this section is to characterise the conditions under which a system transmits a signal faithfully.

4.7.1. Distortionless Transmission

A system transmits $x(t)$ **without distortion** if the output is a scaled, delayed replica:

$$y(t) = Kx(t - t_0), \quad K > 0, \quad t_0 \in \mathbb{R}. \tag{107}$$

Taking the Fourier transform of both sides:

$$Y(j\omega) = Ke^{-j\omega t_0}X(j\omega). \tag{108}$$

Comparing with $Y(j\omega) = H(j\omega)X(j\omega)$, the condition for distortionless transmission is

$$H(j\omega) = Ke^{-j\omega t_0}. \tag{109}$$

This requires:

- **Flat magnitude response:** $|H(j\omega)| = K$ for all ω — every frequency is amplified by the same gain.
- **Linear phase response:** $\angle H(j\omega) = -\omega t_0$ — the phase shifts linearly with frequency.

Any deviation from these conditions introduces distortion. Two types are distinguished:

- **Amplitude distortion:** $|H(j\omega)|$ is not constant — different frequencies are amplified by different amounts, changing the relative sizes of spectral components.
- **Phase distortion:** $\angle H(j\omega)$ is not linear — different frequencies are delayed by different amounts, disrupting the temporal alignment of spectral components.

> **Note**
>
> A system can have a flat magnitude response yet still distort if its phase is nonlinear. Phase distortion is often perceptually significant in audio and communications but invisible to power-spectrum measurements — this is why phase response must always be checked alongside magnitude response.

4.7.2. Group Delay

For a system with phase response $\angle H(j\omega)$, the **group delay** is defined as

$$\tau_g(\omega) = -\frac{\mathrm{d}}{\mathrm{d}\omega}\angle H(j\omega). \tag{110}$$

Group delay measures the time delay experienced by the **envelope** of a narrowband signal centred at frequency ω. It generalises the concept of delay to systems whose phase is not exactly linear.

The distortionless condition $\angle H(j\omega) = -\omega t_0$ is equivalent to **constant group delay** $\tau_{g(\omega)} = t_0$ for all ω: every narrowband component of the signal is delayed by the same amount, so the envelope shape is preserved.

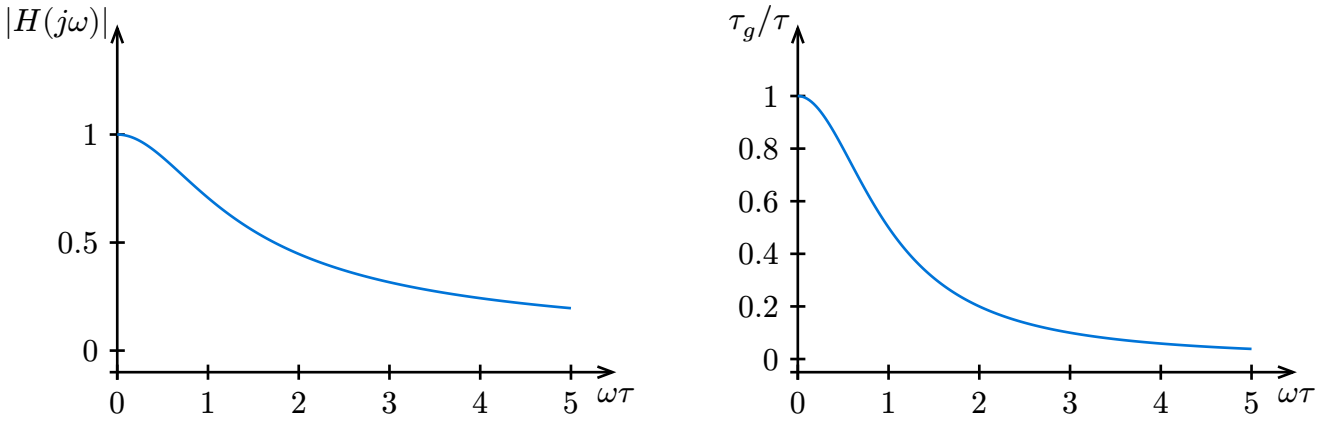
4.7.2.1. Worked Example: RC Lowpass Filter

Returning to the first-order RC filter $H(j\omega) = 1/(1 + j\omega\tau)$, the phase is

$$\angle H(j\omega) = -\arctan(\omega\tau). \tag{111}$$

The group delay is

$$\tau_{g(\omega)} = -\frac{\mathrm{d}}{\mathrm{d}\omega}(-\arctan(\omega\tau)) = \frac{\tau}{1 + (\omega\tau)^2}. \tag{112}$$



The group delay equals τ at DC and falls monotonically toward zero at high frequency. Because $\tau_{g(\omega)}$ is not constant, the RC filter is a **dispersive** system — it delays low-frequency components of the envelope more than high-frequency ones, spreading pulses in time. This is the mechanism behind pulse broadening in dispersive transmission media such as optical fibres.

Key insight

Distortionless transmission requires **linear phase** (equivalently, **constant group delay**). A flat magnitude response alone is not sufficient. Systems with linear phase — such as symmetric FIR filters in the discrete-time setting — are therefore highly desirable whenever waveform shape must be preserved.

4.8. Energy Spectral Density

4.8.1. Definition

From Parseval's theorem the total energy of $x(t)$ is

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, \mathrm{d}t = \text{frac}\,\{1\}\{2\pi\} \int_{-\infty}^{\infty} |X(j\omega)|^2 \, \mathrm{d}\omega. \tag{113}$$

The integrand on the right, $|X(j\omega)|^2 \ /(2\pi)$, describes how the signal's energy is **distributed across frequency**. It is called the **energy spectral density (ESD)**:

$$S_x(\omega) = |X(j\omega)|^2, \tag{114}$$

so that

$$E_x = \text{frac}\,\{1\}\{2\pi\} \int_{-\infty}^{\infty} S_x(\omega) \, \mathrm{d}\omega. \tag{115}$$

$S_{x(\omega)}$ has units of energy per unit angular frequency (J·s/rad). For a real signal $x(t)$, conjugate symmetry gives $|X(-j\omega)| = |X(j\omega)|$, so $S_{x(\omega)}$ is an even function of ω.

4.8.2. Energy in a Frequency Band

The energy contained in the band $\omega_1 \le |\omega| \le \omega_2$ is

$$E_{\omega_1,\omega_2} = \frac{1}{2\pi}\left[\int_{-\omega_2}^{-\omega_1} + \int_{\omega_1}^{\omega_2}\right] S_{x(\omega)}\,\mathrm{d}\omega = \frac{1}{\pi}\int_{\omega_1}^{\omega_2} S_{x(\omega)}\,\mathrm{d}\omega, \tag{116}$$

where the last equality uses the even symmetry of $S_{x(\omega)}$.

4.8.3. Bandwidth Definitions

Several definitions of signal **bandwidth** are in common use, each suited to different contexts:

Definition	Condition
3 dB bandwidth $B_{3\text{dB}}$	$S_{x(\omega)} = S_{x(0)}/2$, i.e. $\lvert X(j\omega)\rvert = \lvert X(0)\rvert/\sqrt{2}$
Null-to-null bandwidth B_{null}	Distance from DC to first spectral null of $\lvert X(j\omega)\rvert$
RMS (Gabor) bandwidth B_{rms}	$B_{\text{rms}}^2 = \frac{1}{E_x}\,\frac{1}{2\pi}\int \omega^2 S_{x(\omega)}\,\mathrm{d}\omega$

Each definition gives the same answer for a Gaussian signal (since Gaussians are their own Fourier transforms) but can differ significantly for other shapes.

4.8.4. Worked Example

Find the ESD, total energy, and 3 dB bandwidth of $x(t) = e^{-at}u(t)$, $a > 0$.

Step 1 — ESD: From §4.3, $X(j\omega) = 1/(a + j\omega)$, so

$$S_{x(\omega)} = |X(j\omega)|^2 = \frac{1}{a^2 + \omega^2}. \tag{117}$$

Step 2 — Total energy:

$$E_x = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{a^2+\omega^2}\,\mathrm{d}\omega. \tag{118}$$

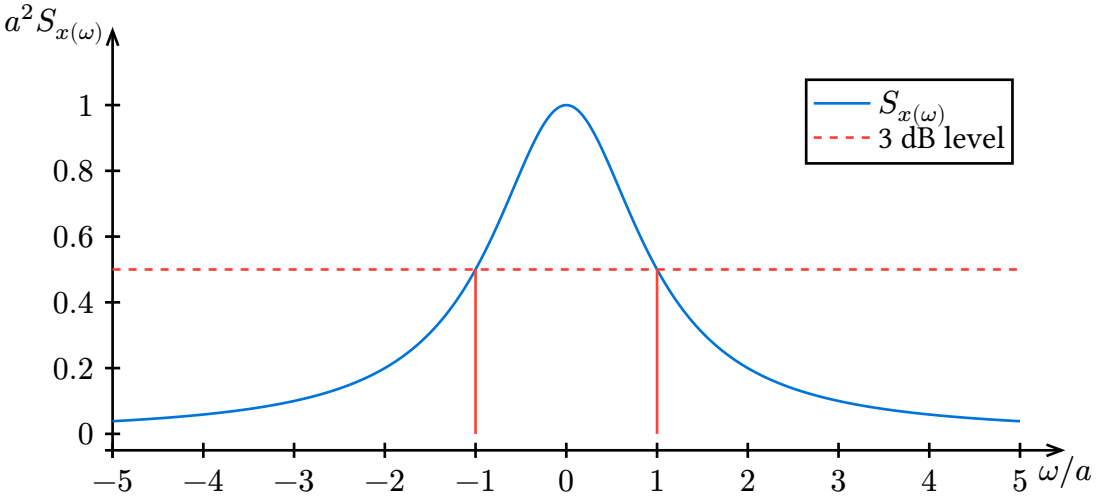
Using the standard result $\int_{-\infty}^{\infty} \mathrm{d}\omega/(\alpha^2+\omega^2) = \pi/\alpha$:

$$E_x = \frac{1}{2\pi}\cdot\frac{\pi}{a} = \frac{1}{2a}. \tag{119}$$

This matches the direct time-domain calculation: $E_x = \int_0^\infty e^{-2at}\,\mathrm{d}t = 1/(2a)$. ✓

Step 3 — 3 dB bandwidth: The ESD at DC is $S_{x(0)} = 1/a^2$. Setting $S_{x(\omega_c)} = S_{x(0)}/2$:

$$\frac{1}{a^2+\omega_c^2} = \frac{1}{2a^2} \quad\Rightarrow\quad \omega_c = a \ \text{ rad/s}. \tag{120}$$



A larger a (faster decay in time) gives a wider ESD ($\omega_c = a$ increases) — again the time–bandwidth trade-off: a signal that is short-lived in time is necessarily broad in frequency.

4.8.5. ESD and the Autocorrelation Function

The ESD is also the Fourier transform of the **autocorrelation function** $R_{x(\tau)}$:

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t)x(t+\tau)\,\mathrm{d}t \overset{\mathcal{F}}{\leftrightarrows} S_{x(\omega)} = |X(j\omega)|^2. \tag{121}$$

This relationship (the **Wiener–Khinchin theorem** for energy signals) is useful when the ESD must be estimated from a signal without direct access to $X(j\omega)$.

5. Butterworth Filter Design

6. Nyquist's Sampling Theorem

7. State-Space Analysis